

What is version control?

Version control system (VCS) keeps a track of the developer's code as it progresses. It maintains a special database in the central repository that stores the modifications made to the code with unique time stamps. In case of mistakes, the developer can retrieve the earlier code from the database without losing much progress. Additionally, it organises the project as a tree structure where each developer confines to a leaf, and the central repository is the root of the tree. This makes it easy to track each developer's work and prevents concurrent work from conflicting as each developer's work is synchronised to the central repository after it is completed. There are many systems which offer these features but we'll be choosing Git because of its popularity and ease of use.

What is Git?

Git is an open source version control system which is currently recognised as de facto standard in software firms. It is available in the form of command line as well as a GUI tool for Linux, Windows and Mac OS X operating systems. Although the GUI version is simpler to use, developers prefer the command-line tool. In addition to all the primary VCS features, Git has lot more to offer in terms of functionality, performance, security, and flexibility. In comparison with the alternatives available, Git's raw performance is better. It uses deep learning algorithm that makes committing, branching, merging, etc lot more optimised than its competitors.

What is GitHub?

GitHub is a Git-powered, web-based repository hosting service which can be accessed through github.com. It hosts millions of open-source software projects from millions of developers across the world. Git maintains version controlling locally on your PC, however, in contrast, GitHub keeps a global copy of your project on the GitHub servers. Keeping a centralised copy of your project on a public server allows for collaborative development for project contributors from across the world. It also makes sharing your ready-to-execute project easy. The foremost aim of making the code open-source is code reusability.

In this section, we will be guiding you for install and managing Git locally on your PC, creating a personal repository on GitHub, and pushing and pulling projects from it.